NOT MEASUREMENT SENSITIVE

MIL-C-50430A(MI)
28 June 1989
SUPERSEDING
MIL-C-50430(MI)
2 June 1970

MILITARY SPECIFICATION

# CLIP, SHORTING ASSEMBLY FOR 2.75 INCH ROCKET MOTOR

(MARK 4 AND 40)

This specification is approved for use within the Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the quality assurance provisions and special requirements not covered by the drawings for one type of shorting clip designated as Clip, Shorting Assembly, for 2.75 Inch Rocket Motor (Mark 4 and 40).

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) at the end of this document or by letter.

AMSC N/A FSC 1340

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:24Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-A-002550 | Ammunition, General Specification for |
| MIL-I-45607 | Inspection Equipment, Acquisition, Maintenance and Disposition of |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-376 | Preferred Metric Units for General Use by the Federal Government |

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | Quality Assurance Terms and Definitions |
| MIL-STD-1168 | Ammunition, Lot Numbering |
| MIL-STD-1169 | Packaging, Packing and Marking for Shipment of Inert Ammunition Components |
| MIL-STD-1235 | Single and Multi-Level Continuous Sampling Procedures and Tables for Inspection by Attributes Functional Curves of the Continuous Sampling Plans |

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

| | |
|---|---|
| 9242646 | Shorting Clip Assembly |

PUBLICATIONS

ARMY

| | |
|---|---|
| ET-9242646 | Equipment Tabulation |

(Copies of drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:24Z
Check the source to verify that this is the current version before use.

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The materials and parts shall be as specified herein and in accordance with drawings 9242646 and ET-9242646.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.3 Workmanship. Clips shall meet all design dimensions and intermateability requirements of this specification. They shall be free of all foreign matter. Metal parts shall be free of rust and other foreign matter. Any cleaning methods used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agent. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Submission of product. At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:24Z
Check the source to verify that this is the current version before use.

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing, specification number and date, together with identification and date of changes.

c. A statement that all material purchased by the contractor meet requirements. When such material is controlled by Government or commercial specifications referenced in any of the contractual documents, a statement that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in applicable test method document or applicable paragraph(s) in the specification.

4.4 First article inspection.

4.4.1 Submission. The contractor shall submit a first article sample to a Government approved facility designated by the contracting activity (see 6.3) for evaluation in accordance with 4.4.2. The sample shall consist of 30 shorting clip assemblies and 30 of each part which have been produced by the contractor or manufactured by a supplier using the same production processes, procedures and equipment including packaging and packing, and obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings and certificates of conformance for materials (see 4.1.2c). A sample containing known defects will not be submitted unless specifically authorized by the contracting activity. A first article sample or portion thereof, as directed by the



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:24Z
Check the source to verify that this is the current version before use.

contracting activity shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in the manufacturing process, material used, drawings or specifications such as to significantly affect product uniformity as determined by the Government.

4.4.2 Inspections to be performed. Parts will be subjected by the Government to any or all of the examinations or tests specified in this specification and to drawing requirements.

4.4.3 Noncompliance. If any part fails to comply with any of the requirements and inspections, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a part in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and/or submit a new first article sample or portion thereof. Until a first article sample is accepted the contractor is in no way authorized by the Government to initiate regular production unless directed by the Contracting Officer.

4.5 Quality conformance inspection.

4.5.1 Lot formation. The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of production from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-109. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall bear a unique lot number.

4.5.2 Examination. The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.5.1) of the components or assemblies in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the acceptable quality level (AQL's) designated herein to the groups of major and minor defects listed, the contractor may elect, to apply the AQL's to individual defects using an AQL of 0.40 percent for each major defect and an



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:24Z
Check the source to verify that this is the current version before use.

AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspection listed shall be in accordance with 4.5.4.

4.5.2.1 Shorting clip assembly (see drawing 9242646).

| CATEGORIES | DEFECTS | METHOD OF INSPECTION | CODE NO. (See 6.) |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 1.0 percent | | |
| 101. | Width of loop ..... | Gage | 01001 |
| 102. | Depth of loop ..... | Gage | 01002 |
| 103. | Length of side leg (1.50 nominal) ... | Gage | 01003 |
| 104. | Distance from side leg to center of of ripple (1.40 nominal) ..... | Gage | 01004 |
| 105. | Distance of offset in cross leg (.32 nominal) ..... | Gage | 01005 |
| 106. | Spring clip loose ..... | Visual/Manual | 01006 |
| 107. | Evidence of poor workmanship (see 3.2) ..... | Visual | 01007 |
| Minor: | None defined. | | |

4.5.3 Testing.

4.5.3.1 Disassembly force test - major defect - code No. 01008 (see drawing 9242642 Note 8). A sampling plan according to the schedule in table I shall be used for each lot of shorting clip assemblies. Any shorting clip which fails to disassemble with the required force when tested as specified in 4.6.1 using equipment in accordance with 4.5.4 shall be considered defective. This test is considered destructive and units tested shall not be returned to the lot.

TABLE I. Sampling schedule.

| Lot Size | Sample Size | Accept | Reject |
|---|---|---|---|
| to 50,000 | 125 | 1 | 2 |
| 50,0001 to 100,000 | 200 | 2 | 3 |
| 100,001 to 200,000 | 300 | 3 | 4 |
| 200,001 and over | 450 | 4 | 5 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:24Z
Check the source to verify that this is the current version before use.

4.5.4 Inspection equipment. Equipment Tabulation (ET) number ET-9242646 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.5.4.1 Government rights to documentation. Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.6 Test methods.

4.6.1 Disassembly force test - (see 4.5.3.1). Each shorting clip assembly in the sample shall be assembled to a test fixture as specified on the drawing. A load shall be applied at a uniform rate from zero toward 25 pounds (11.35 kg) [1/]. Observation shall be made of the load at disassembly.

4.7 Inspection of packaging. The sampling and inspection of the preservation, packaging, and container marking shall be as specified (see 5.1.1).

5. PACKAGING

5.1 Packing and marking.

5.1.1 Level C. The shorting clips shall be packed and marked in accordance with MIL-STD-1169.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The shorting clip assembly covered by this specification is intended for use with the 2.75 Inch Rocket Motor, (Mark 4 and 40).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of the DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

---

[1/] Wherever inch/pound dimensions are used in this document, metric equivalents in accordance with FED-STD-376 shall be acceptable.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:24Z
Check the source to verify that this is the current version before use.

c. Whether a first article sample is required (see 3.2 and 4.4).

d. Quality required.

e. Levels of preservation and packing (see section 5).

6.3 First article. When a first article inspection is required, the items should be a first article sample (see 4.4.1). The contracting officer should include specific instructions in acquisition documents regarding arrangement for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or tests, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Inspection code numbers. The five-digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government (see 4.5.2.1). These code numbers are also used to correlate the characteristics cited on equipment lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval.

6.5 Inspection equipment designs. Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers in the paragraph headings in section 4. Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD." The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.5.2 and 6.5.3 should he elect to do so.

6.5.1 SIE designs. SIE designs may consist of any of the following:

a. Detailed drawing which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing.

c. An envelope drawing which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.5.2 Contractor designs. Contractor designs are required for all inspection equipment for which SIE designs are not specified, and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is catalogued and available



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:24Z
Check the source to verify that this is the current version before use.

for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government, and may include illustrations and engineering data. Designs shall be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.5.1c). Designs shall be of the category and form specified in the Contract Data Requirement List (DD 1423). The specification number and the applicable five-digit code number (or other specific identifying information) from section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

6.5.3 Submission of designs for approval. Contractor design shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirement List (DD 1423) or in the contract. Partial submission of the inspection equipment designs is permissible and encouraged. However, the complete date for design review will be based on the date of the final submission of designs.

6.6 Subject term (keyword) listings.
Assembly
Clip
Motor
Rocket
Shorting

6.7 Changes from previous issue. Asterisks or vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Preparing Activity:
Army - MI
Project No. 1340-A748



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:24Z
Check the source to verify that this is the current version before use.